

Arin Balyan

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Experience

BrilliED | Full-Stack AI Engineer

Nov 2025 – Present

FastAPI, Python, Vue.js, LangChain, PostgreSQL, Supabase, Docker, CI/CD

Remote — Full-time

- Architected and sole-engineered brillied.in end-to-end, serving **1,000+** daily users with **99.8% uptime** across 12 production releases with full DevOps ownership.
- Built **RAG pipeline** using LangChain generating **500+** study pages and **100+** quizzes at **92% accuracy**, cutting content creation time by **85%** via automated workflows.
- Optimized DB queries and caching, reducing API response time by **77%** (800ms → 180ms) and absorbing **300% traffic growth** at zero additional infrastructure cost.
- Established CI/CD pipeline with **85% test coverage**, slashing deployment time from 45 min to 5 min; mentored 2 developers improving code quality by 40%.

SkillBit Technologies | AI Intern

Jul 2025 – Aug 2025

Python, FastAPI, Scikit-learn, Pandas, NLP

Remote

- Built NLP-based Sentiment Analysis pipeline for customer feedback using Scikit-learn and Pandas; deployed as **FastAPI** REST endpoints integrated into production workflow.
- Collaborated in agile team on data preprocessing pipelines, improving model input quality and reducing noise in training datasets.

Projects

EmotionML | Real-Time Emotion Recognition Platform

Jul 2025 – Aug 2025

PyTorch, MobileNetV2, FastAPI, React.js, OpenCV

[GitHub](#)

- Trained **9 deep learning models** on **35,000+** images achieving **98.99% accuracy** on CK+48, outperforming baselines by 12% via hyperparameter tuning.
- Optimized MobileNetV2 reducing model size **60%** (23MB → 9MB) and inference time **40%** (250ms → 150ms), enabling edge deployment on 512MB RAM devices.
- Built FastAPI backend processing **1,000+** images/hour with GPU acceleration (10 → 30 req/sec), supporting **50+** concurrent users.

Image Enhancement Web Service | AI Photo Processing Pipeline

Jan 2026 – Present

Python, FastAPI, PyTorch, Real-ESRGAN, GFPGAN, OpenCV

[GitHub](#)

- Engineered production-ready service combining **Real-ESRGAN** (4× super-resolution), **GFPGAN** face restoration, and adaptive white-balance/exposure correction.
- Built automated image analysis module to intelligently detect and apply required enhancements; supports **batch processing** and **Google Drive** integration via FastAPI.

AgroAid | ML-Powered Agricultural Assistant

Nov 2024 – Feb 2025

Python, TensorFlow, XGBoost, Flask, NLP

[GitHub](#)

- Built ensemble model (Random Forest + XGBoost + stacking meta-learner) on **50,000+** records from 3 government datasets achieving **89% crop yield accuracy**, outperforming single models by 15 pp.
- Engineered multilingual recommendation engine in **5 languages** (Hindi, English, Tamil, Telugu, Kannada) serving **500+** daily queries with **99.5% uptime**.

Technical Skills

Languages: Python, C++, JavaScript, TypeScript, SQL

AI / ML: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Hugging Face

LLM / Agentic AI: LangChain, RAG Systems, Prompt Engineering

DevOps / Cloud: Docker, AWS, CI/CD, Git, Linux, Render, Cloudflare

Backend: FastAPI, Flask, Node.js, REST APIs

Databases: PostgreSQL, MySQL, Redis, Supabase

Frontend: React.js, Vue.js, Next.js, Tailwind CSS

Tools: Kaggle, Streamlit, Postman, Selenium

Education

Bachelor of Technology — Computer Science (AI & ML Specialization)

Aug 2022 – 2026

Vellore Institute of Technology

CGPA: 8.37 / 10

Courses & Certifications

[Applied Machine Learning in Python](#) — University of Michigan, Coursera

Score: 94%

[MERN Full-Stack Developer Certification](#) — Ethnus

Score: 90%

[IBM Blockchain Developer Certification](#) — IBM

Achievements & Leadership

- Top 25 Global** among 500+ teams — **Zelestria × AWS Hackathon**: built and deployed an ML solution end-to-end in 48 hours.
- Top 10** among 1,500+ participants — **IIM Nagpur's Arthneeti 4.0**: led 4-member team in market analysis and strategic presentation.
- GSSoC Open-Source Contributor**: 50+ merged PRs across 15+ repositories, collaborating with 100+ developers with 30+ positive code reviews.